Installation Guide: FF_VA Mod for the Original Famicom

This mod improves video signal quality and sound. Before you begin, make sure the console is in working order.

1. Preparing the mod board (jumper and component configuration)

- **Assembly:** I recommend starting with the SMD components, then the through-hole capacitors and larger parts, then the connectors.
- **Power:** different supporting components are soldered depending on the voltage regulator used.
  - For the adjustable LM2596-ADJ: solder R9, R10 and C6 as shown in the schematic.
  - For the fixed LM2596-5.0: do not solder R10 and C6, and replace R9 with a 0 Ohm jumper.
- **Video_OUT signal jumpers** (depending on which BA7622 channel is used):
  - **IN1** – solder (close) **JP1** and **C3**. Do **not** solder (leave open) **JP2** and **C4**.
  - **IN2** – do **not** solder (leave open) **JP1** and **C3**. Solder (close) **JP2** and **C4**.
- **Diagnostic elements (JP3 and C5):**
  - Needed only to test the console with its stock video output after the mod has been connected for power and VA signals.
  - **Recommendation:** if the console is known to be working, **do not solder C5 and do not close JP3**. If you soldered them for testing, be sure to desolder C5 and open JP3 after the test, before final assembly.

2. Connecting power and VA signals

- **Main power:** connect the console's main board to the mod through connector **J5** or **J4** (choose whichever is more conveniently placed in the case), following the markings on the mod board.
- **Power button:** extend the cable from the stock button and connect it to connector **J2 (PWR_sw)** on the mod board.

3. Connecting video (PPU)

- Take the video signal directly from **pin 21** of the PPU.
- Connect it to contact **J7 (Video_21_in)**.
- **Important:** use **shielded cable** to avoid interference and a "grid" pattern on the picture.

4. Audio modification

For clean, noise-free sound, follow these steps:

1. **Cartridge slot:** carefully cut the trace that runs to **pin 45** of the slot.
2. **CPU:** desolder the resistor dividers on CPU outputs **1 and 2** (the area boxed in red in the figure below).
3. **Wiring:** connect **CPU pins 1 and 2** to the corresponding inputs on the mod board.

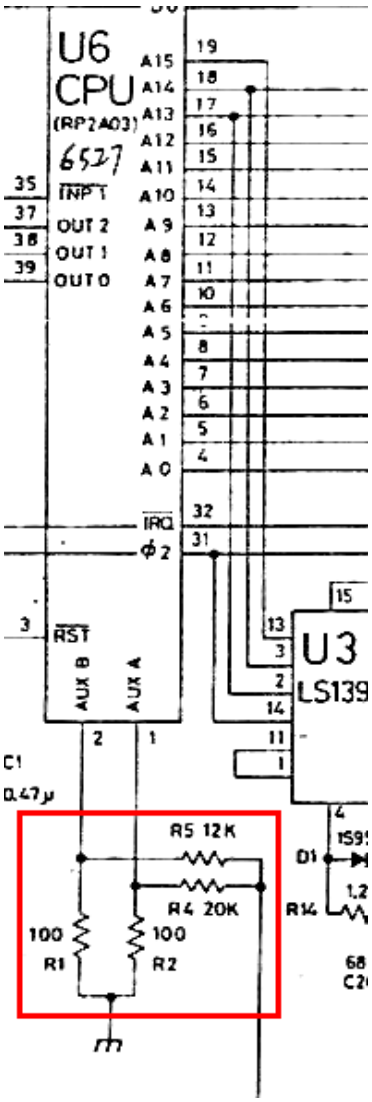


Figure: CPU (U6) audio outputs AUX A / AUX B with the resistor dividers (R1, R2, R4, R5) to be removed.

- **Tip:** shielded wire is also strongly recommended here.

5. Important assembly note (LM2596)

If you are using the LM2596 regulator in a **through-hole package**:

- Before installing it, cover the SMD pads underneath with an insulating **mask**.
- *Reason:* the chip's heatsink tab (ground) runs very close to the signal traces, which can cause a short circuit if it is deformed or pressed tightly into place.